Troubleshooting Guide: Molecular Sieve Dehydration — Early Water Breakthrough

Category: Gas Processing / Solid Adsorption **Unit:** Molecular Sieve Dryer Beds (TSA cycle), upstream of NGL/cryogenic processing **Tools:** Cyclic adsorption performance/breakthrough-curve analysis, regeneration temperature trending **Fluid Package:** Not applicable in the VLE sense — performance is governed by adsorption isotherms and regeneration thermal profiles, not a phase-equilibrium flash **Symptom:** Outlet gas dew point creeping up before the normal end of the adsorption cycle — beds "breaking through" early

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Note on case type: Like the seawater cooler case (Case Study 6), this is fundamentally about **cyclic performance trending**, not phase-equilibrium simulation. Adsorption capacity, not VLE, is the governing physics — the diagnostic approach centers on regeneration effectiveness, not a fluid package choice.

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1. Quick Reference

Item	Detail
Reported symptom	Outlet gas dew point rising noticeably before scheduled bed switchover
Initially unclear	Whether beds were simply reaching end-of-life, or something in the cycle itself had degraded
Actual root cause	Regeneration gas heater underperforming — regen gas not reaching design temperature, leaving residual moisture on the sieve each cycle and progressively reducing dynamic capacity
Fix	Corrected heater duty/instrumentation issue; verified regen outlet temperature reached design target
Diagnostic signal	Bed differential temperature profile during regeneration not reaching expected peak; lab analysis of pulled sieve confirmed retained moisture
Prevention	Trend regeneration outlet temperature every cycle; alarm if target not reached

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2. Symptom

- **Outlet gas dew point began rising before the normal end of the adsorption cycle**, i.e., beds appeared to be breaking through earlier than their design cycle time.
- No obvious upstream process change (feed rate, inlet moisture loading) was reported.

3. Why This Wasn't Simply "Beds Need Replacing"

Early breakthrough is often assumed to mean the sieve has reached end-of-life and needs replacement — an expensive assumption to act on without confirmation. But early breakthrough can equally be caused by **incomplete regeneration** each cycle: if a bed isn't fully dried out before being put back into adsorption service, its *effective* capacity shrinks even though the sieve material itself may still be perfectly good. Distinguishing these two requires looking at the **regeneration side** of the cycle, not just the adsorption-side symptom.

4. Diagnostic Approach

Step 1 — Review the adsorption-side breakthrough trend

Outlet dew point trends confirmed breakthrough was occurring measurably earlier than the design cycle time, and that this had developed progressively (not a single-cycle anomaly).

Step 2 — Check the regeneration-side temperature profile

Bed differential temperature during regeneration was reviewed cycle-by-cycle. **The regeneration outlet temperature was not reaching its design target** — the bed was being taken off regeneration and put back into service before it was fully dried.

Step 3 — Confirm physically with a sieve sample

A sieve sample pulled from an affected bed was analyzed in the lab and **confirmed retained moisture** consistent with incomplete regeneration — not sieve degradation (attrition, crushing, contamination) that would indicate genuine end-of-life.

Step 4 — Trace back to the regeneration heater

With incomplete regeneration confirmed as the mechanism, the investigation moved to *why* — the regeneration gas heater was underperforming, not delivering enough duty to reach design regen temperature within the allotted cycle time.

Quantitative Basis

- Design adsorption cycle time: 8 hr, with an outlet dew point spec of ≤ −40°F. Breakthrough was occurring at **6.1 hr — 24% early.**
- Design regeneration outlet temperature target: 550°F. Trended actual regen outlet averaged **495°F — 55°F below target**, meaning beds were returned to service before reaching design dryness.
- Lab analysis of a pulled sieve sample measured **4.8 wt% retained moisture, versus <0.5 wt% for a properly regenerated bed** — direct confirmation of incomplete drying rather than sieve attrition or contamination.

5. Root Cause

The regeneration gas heater was underperforming, so regeneration gas did not reach the design outlet temperature within the cycle time. This left **residual moisture on the sieve** at the start of each new adsorption cycle, **progressively reducing dynamic capacity** and causing water to break through earlier and earlier as the effect compounded cycle over cycle.

6. Corrective Action

1. Corrected the heater duty/instrumentation issue causing underperformance.
2. Verified regeneration outlet temperature reached its design target on subsequent cycles.

7. Verification

- Regeneration outlet temperature restored to an average of **552°F**, at/above the 550°F design target.
- Breakthrough timing returned to **7.9 hr**, close to the 8 hr design cycle time.
- Outlet dew point held below **−42°F over 30 days (≈90 cycles)** of continued monitoring.

8. Prevention / Long-Term Fix

- **Regeneration outlet temperature is now trended every cycle**, with an alarm if the design target is not reached — catching incomplete regeneration immediately rather than after several cycles of cumulative capacity loss.

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9. General Troubleshooting Checklist (Reusable)

- ☐ Confirm breakthrough timing against the design cycle time using outlet dew point trend
- ☐ Before assuming end-of-life, review the **regeneration-side** temperature profile for each affected bed
- ☐ Confirm regeneration outlet temperature is reaching its design target, not just that the heater “ran”
- ☐ Pull and lab-analyze a sieve sample to distinguish incomplete regeneration (retained moisture) from genuine sieve degradation (attrition, crushing, coking, heavy hydrocarbon contamination)
- ☐ If regeneration is incomplete, trace back to the heater (duty, instrumentation, fouling) or cycle timing logic
- ☐ After correction, confirm both regeneration temperature profile AND adsorption-side breakthrough timing return to design
- ☐ Add per-cycle regeneration temperature trending/alarming to standing operations monitoring

10. Key Takeaway

> Early breakthrough on a cyclic adsorption system isn’t automatically an end-of-life sieve problem — it can just as easily be an **incomplete regeneration** problem that silently compounds cycle after cycle. Before replacing expensive sieve material, confirm the regeneration side is actually delivering its design temperature and duration; a lab sample of the pulled sieve is the cheapest way to tell “needs replacing” from “isn’t being dried properly” apart.

Related Concepts / Tags

`molecular-sieve` `TSA` `dehydration` `breakthrough-curve` `regeneration-temperature` `adsorption-capacity` `dew-point` `cyclic-process` `gas-processing`

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.